

Classical Mechanics

Homework Assignment #1 (40 pts)

(Due on 11 Sep 2026)

Problem 1 (4 pts)

The averaged matter density of the Universe is $4.21 \times 10^{-28} \text{ kg/m}^3$. The diameter of the observable universe is 93 billion light-years. Given the above number, estimate

- (a) how many protons are within the observable Universe?
- (b) how many stars are within the observable Universe?

Proton mass = $1.67 \times 10^{-27} \text{ kg}$, stellar mass = $1.99 \times 10^{30} \text{ kg}$.

Problem 2

Prove following vector identities

- (a) (8 pts)

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C} \quad (1)$$

- (b) (6 pts)

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \quad (2)$$

- (c) (4 pts)

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) + \mathbf{B} \times (\mathbf{C} \times \mathbf{A}) + \mathbf{C} \times (\mathbf{A} \times \mathbf{B}) = 0 \quad (3)$$

- (d) (8 pts)

$$(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{C} \times \mathbf{D}) = (\mathbf{A} \cdot \mathbf{C})(\mathbf{B} \cdot \mathbf{D}) - (\mathbf{A} \cdot \mathbf{D})(\mathbf{B} \cdot \mathbf{C}) \quad (4)$$

Problem 3 (4 pts)

In 2D, a point can be described by the radial direction r and the transverse direction θ (instead of x and y). The unit vector along the radial and transverse direction are denoted by $\hat{r}$ and $\hat{\theta}$, respectively. These are related to the Cartesian unit vectors $\hat{i}$ and $\hat{j}$ by

$$\hat{r} = \cos \theta \hat{i} + \sin \theta \hat{j}, \quad \hat{\theta} = -\sin \theta \hat{i} + \cos \theta \hat{j}. \quad (5)$$

Show that

- (a) $|\hat{r}| = |\hat{\theta}| = 1$;
- (b) $\hat{r} \perp \hat{\theta}$;
- (c) $d\hat{r}/d\theta = \hat{\theta}$, $d\hat{\theta}/d\theta = -\hat{r}$.

Problem 4 (6 pts)

If $\mathbf{a} = \mathbf{a}(t)$ and $\mathbf{b}$ is a constant vector, show that

$$\frac{d}{dt} \left[\mathbf{a} \cdot \left(\frac{d\mathbf{a}}{dt} \times \mathbf{b} \right) \right] = \mathbf{a} \cdot \left(\frac{d^2\mathbf{a}}{dt^2} \times \mathbf{b} \right). \quad (6)$$